

TO SUPPLY LEFTOVER FOOD TO POOR

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Team ID	LTVIP2025TMID29611
Project Name	To Supply Leftover food to Poor
Maximum Marks	

Chapter 4

Project Design

4.1: Existing System (Problem Solution Fit)

Problem	Solution
Surplus food is wasted daily	Donor portals to list available food
No coordination with NGOs	Real-time matching and alerts
Manual tracking and delays	Automated pickup scheduling and routing
Food safety concerns	Hygiene checklists and expiry tracking
Lack of impact visibility	Dashboards and analytics

4.2: Proposed System (Proposed solution)

Concept Summary

The system is a **real-time web application** that connects food donors (restaurants, event venues, households) with NGOs and volunteers who distribute the food to those in need. It aims to reduce food waste and alleviate hunger simultaneously.

System Modules

Module	Description
Donor Portal	Donors register, list surplus food, specify pickup time and location
NGO Portal	NGOs browse listings, claim food, and coordinate pickups
Logistics	Delivery agents manage transportation and route optimization
Admin Panel	Verifies users, maps donors to NGOs, monitors feedback and system health

Key Features

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- **Real-Time Notifications:** Alerts NGOs when new food listings are available
- **Food Tracking System:** Monitors food from listing to delivery
- **Secure Authentication:** Ensures only verified users participate
- **Impact Dashboard:** Tracks donations, beneficiaries, and food saved
- **Feedback Mechanism:** Users can report issues or suggest improvements

Technologies Used

- **Platform:** Web-based or Salesforce CRM
- **Tools:** Apex, Lightning Components, Workflow Rules
- **Database:** Cloud-based object storage
- **Communication:** SMS and email alerts

4.3: Solution Architecture

Component	Role
Donor Portal	Allows restaurants, event venues, and individuals to list surplus food
NGO Portal	Enables NGOs to browse, claim, and track food donations
Web Server	Hosts the application and handles API requests
Admin Panel	Manages user verification, system health, and feedback
Logistics Module	Assigns delivery agents, optimizes routes, and confirms deliveries
Analytics Engine	Tracks food saved, meals served, and environmental impact
Notification Hub	Sends real-time alerts via SMS/email
Cloud Database	Stores user data, food listings, and transaction logs securely

Benefits of This Architecture:

- **Scalable:** Easily handles growing donor and NGO base
- **Secure:** Protects sensitive user and food data
- **Real-Time:** Ensures timely pickups and minimal food spoilage
- **Impact-Driven:** Tracks metrics for social and environmental reporting